

# Supplementary Material for “Conflict-Complete Retrieval: Checkable Contradictions under a Context Budget”

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## Scope

This supplement expands proofs and records the reviewer-facing reproduction contract. It does not introduce an additional headline claim or repair an argument missing from the main paper. The main paper remains self-contained. Core results concern deterministic synthetic fixtures; the semi-natural leg preserves public gold linkage topology and natural attributes but injects the single-valued conflict predicate. Neither supplies a natural-prevalence estimate or independently adjudicated semantic conflict labels.

## Assumption Ledger

The guarantees use the following assumptions.

1. The premise store is finite after applying the confidence threshold. Assertions have exact RDF-term identity and retain their source identifiers.
2. Each keyed class has one supplied key predicate. Key equality is exact; no entity linker, string normalization, or probabilistic match is inferred.
3. Only schema-declared single-valued predicates can generate the conflicts studied here. Time intervals must overlap and scopes must agree unless an annotation is explicitly unspecified.
4. Query relevance is determined by supplied entity seeds and requested predicates. Conflict completeness requires one complete minimal witness for every relevant conflict, not every alternative derivation. Context cost is monotone under set inclusion.
5. The maintained index is evaluated on the current insertion-oriented snapshot. Deletion, persistence, concurrent mutation, and crash recovery are outside the implementation contract.
6. Exact joint selection enumerates simple identity paths and searches their witness combinations. Path and search-node caps can force safe SEARCH-LIMIT; only an exhaustive run claims optimality or infeasibility.

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## Expanded Correctness Arguments

### Detector Soundness and Completeness

Let direct co-keying connect two typed records that share the exact key value, and let its reflexive, symmetric, transitive closure be  $\sim_K$ . Propagation copies only schema-constrained values within one  $\sim_K$  class. Therefore a derived unequal-value pair has endpoints in one semantic entity class and satisfies the predicate, confidence, time, and scope guards. This proves soundness.

For completeness, take any semantic conflict. Its endpoint records are in one finite  $\sim_K$  class, so a finite path of direct co-key steps connects them. Repeated propagation along the path carries both constrained values to a common representative, where the unequal-value rule fires. The finite source leaf set of that derivation contains a minimal subset: repeatedly delete a leaf whenever entailment remains. Termination is immediate because the leaf set is finite. The remaining subset is a witness by definition.

### Joint Budget Feasibility

For a finite relevant conflict set  $\Gamma$ , choose one witness  $W_\gamma \in \mathcal{W}_\gamma$  for each conflict and take their union. Finiteness of the premise store makes the collection of possible unions finite, so a minimum-cost union  $U^*$  exists. If  $\mu(U^*) \leq B$ , returning  $U^*$  is conflict-complete and respects the budget. Conversely, any conflict-complete context  $R$  contains some witness  $W_\gamma$  for every  $\gamma$ . Their union is a subset of  $R$ ; monotonicity gives  $b^* \leq \mu(R) \leq B$ . This proves the if-and-only-if boundary.

Let  $\widehat{W}_\gamma$  be the shortest witness independently selected by CWI and  $\widehat{U} = \bigcup_\gamma \widehat{W}_\gamma$ . Under triple count, each  $\widehat{W}_\gamma$  is no larger than the witness for  $\gamma$  contained in  $U^*$ , and every such witness is no larger than  $U^*$ . Hence

$$|\widehat{U}| \leq \sum_{\gamma \in \Gamma} |\widehat{W}_\gamma| \leq |\Gamma| |U^*| = |\Gamma| b^*.$$

The implementation deduplicates  $\widehat{U}$  before testing the budget. It returns the entire union if it fits; otherwise it returns SELECTION-OVERFLOW and an empty context. The qualifier “selection” is essential because a different combination of non-shortest witnesses could have a smaller union.

For completeness, the joint-selection decision problem is in NP when witness families are explicit: a certificate

chooses one witness per conflict, and its union and size are polynomially checkable. For NP-hardness, reduce Set Cover with universe  $E$  and candidate sets  $A_1, \dots, A_m$ . Create conflict  $\gamma_e$  and private token  $d_e$  for every  $e \in E$ , shared token  $a_j$  for every candidate set, and witness family  $\mathcal{W}_{\gamma_e} = \{\{d_e, a_j\} : e \in A_j\}$ . Every joint union contains the  $|E|$  private tokens. Its remaining tokens name candidate sets, and choosing one available witness per element is exactly choosing sets that cover  $E$ . Thus  $b^* \leq |E| + k$  if and only if a cover of size at most  $k$  exists.

### Exact Joint Selection

Represent a direct co-key assertion pair as a record–bucket–record edge. Any minimal identity proof connecting two conflicting endpoint records can contain no cycle: deleting a cycle edge preserves endpoint connectivity and contradicts minimality. It therefore corresponds to a simple path in the bipartite record–bucket graph. Conversely, the type and key premises on any simple path, together with the endpoint values and required annotations, form an inclusion-minimal witness: removing any path premise disconnects that certificate, and removing an endpoint value removes the violation.

The exact selector depth-first enumerates these paths, deduplicates equal source-triple sets, and performs branch-and-bound over one witness choice per conflict. The default caps are 10,000 paths per conflict and 1,000,000 search nodes. The independently shortest union initializes the upper bound. When enumeration and selection finish, every tuple in the finite Cartesian product  $\prod_{\gamma \in \Gamma} \mathcal{W}_\gamma$  has either been visited or pruned because its current union is already no smaller than a complete incumbent. The incumbent therefore has size  $b^*$ . A fitting incumbent is safe even if a cap is reached; a non-fitting incumbent justifies INFEASIBLE only after exhaustive search. Otherwise the implementation returns SEARCH-LIMIT and an empty context.

### Fixed-Hop IRI Incompleteness

Fix any finite hop limit  $k$ . Construct typed records  $x$  and  $y$  whose key objects are the same literal, whose constrained values disagree, and for which no triple has an IRI-object path between  $x$  and  $y$ . A query seeded at  $x$  has a relevant direct-key conflict because exact literal-key equality places the records in one  $\sim_K$  class. Its logical witness contains the two type, two key, and two value assertions.

The IRI-only graph omits literals as vertices. Consequently  $x$  and  $y$  are in different connected components, not merely more than  $k$  hops apart. Any policy restricted to the  $k$ -hop component of  $x$  omits the type, key, and value premises incident to  $y$  and cannot contain the witness. Since the construction works for every finite  $k$ , no fixed finite radius gives the claimed guarantee for this policy class. Literal-aware and key-aware policies are outside the proposition.

### CWI Contract and Witness Minimality

After all assertions in a snapshot have been processed, a record belongs to a key bucket exactly when its type and key premises are present. Union-find therefore represents precisely the connected components of the record–bucket

| Family   | Instances | Conflicts | Mechanism                |
|----------|-----------|-----------|--------------------------|
| D20      | 14        | 8         | same/direct key          |
| XR-small | 16        | 12        | direct key + distractors |
| XR-scale | 64        | 60        | scaled direct key        |
| Chain-2  | 41        | 10        | two co-key links         |
| Chain-3  | 25        | 6         | three co-key links       |
| H3-NK1   | 16        | 12        | hierarchy/non-key        |
| H3-NK3   | 16        | 12        | more non-key predicates  |
| TauSig   | 60        | 12        | time/scope guards        |
| Total    | 252       | 132       |                          |

Table 1: The deterministic core synthetic fixture matrix.

graph, which are the  $\sim_K$  classes. Within each class, the index compares only schema-constrained values and applies the same time/scope guard as the formal semantics. Hence its materialized conflict set is sound and complete for the finite snapshot.

For a reported cross-record conflict, breadth-first search selects a shortest record–bucket path between the two endpoint records. The certificate contains both endpoint values and, for every path edge, the type and key premises that justify the edge. Removing either endpoint value removes the unequal-value premise. Removing any path premise breaks that co-key edge; because the path is shortest and the certificate contains no alternative path, the endpoints are no longer connected inside the certificate. Thus no proper subset entails the conflict. A chain of  $m$  direct co-key links has  $m + 1$  type assertions,  $2m$  key assertions, and two endpoint values, for  $3m + 3$  logical assertions. A same-record conflict needs only the two unequal values.

Insertion order does not change the final snapshot relation. A record enters a bucket only once its required type and key are both known; a later premise triggers the same membership update that an earlier premise would have triggered. Later constrained values and annotations re-evaluate their touched class. The tests exercise permutations in which types, keys, values, and annotations arrive before and after one another.

For a multi-seed query, the budget wrapper canonicalizes conflicts reached from more than one seed, filters the requested predicate footprint, and deduplicates source assertions by stable triple identifier. Focused tests verify shared-premise deduplication, exact-budget success, one-below-budget overflow with no partial context, predicate filtering, alternative-path enumeration, exact union minimization, proven infeasibility, and capped-search abstention.

### Evaluation Matrix

For every instance, the shipped manifest records the query, expected label, fixture identifier, and stable source data. Exact and maintained systems are scored by a checker over the assertions they return. A positive counts only when the returned context contains a complete source-grounded witness. Negative controls include different entities, shared non-key values, disjoint time intervals, and unequal scopes.

The retrieval table in the main paper selects the small-

| $g$ | Shared sum | Shared $b^*$ | Disjoint $b^*$ |
|-----|------------|--------------|----------------|
| 1   | 6          | 6            | 6              |
| 2   | 12         | 8            | 12             |
| 4   | 24         | 12           | 24             |
| 8   | 48         | 20           | 48             |
| 16  | 96         | 36           | 96             |
| 32  | 192        | 68           | 192            |

Table 2: Exact triple costs in the unique-path joint-budget fixtures.

est tested configuration that reaches complete recall within a family. The tested record-level depths are 1–4; relevance methods use  $K \in \{1, 2, 4, \dots, 512\}$ . Reciprocal-rank fusion uses 60. Means and maxima are over conflict queries. The scale companion uses one unrecorded warm-up and 11 measured repetitions and reports medians and interquartile ranges. These are descriptive system measurements, not statistical-significance claims.

### Multi-Conflict Budget Matrix

The shared family places  $g$  conflicting predicates on the same two directly co-keyed records. Its four type/key premises are shared and each conflict adds two values, giving  $b^* = 4 + 2g$ . The disjoint family uses  $g$  separate record pairs and multiple query seeds, giving  $b^* = 6g$ . For every row, the shipped runner checks both  $B = b^*$  and  $B = b^* - 1$ .

The alternative-path fixture contains four conflicts. Each has a private two-link shortest path and four alternatives, one of which traverses a shared three-link spine. Independent shortest selection uses 33 triples. Exhaustive selection over 20 candidates visits 226 search nodes, proves a 27-triple optimum, returns complete evidence at  $B = 27$ , and proves infeasibility at  $B = 26$ . Capping each family at one enumerated path instead yields SEARCH-LIMIT, not a false infeasibility claim.

### Semi-Natural External Matrix

The external runner reads `tableA.csv`, `tableB.csv`, and `matches.csv` from the DBLP–Google Scholar entity-resolution benchmark, verifies their recorded SHA-256 digests, and constructs components from supplied gold match edges. Selection is lexicographic and component-disjoint: rare depth-four components are reserved first, followed by depths three, two, and one, with a target of 20 pairs per depth. The available data yield 20 pairs at depths 1–3 and eight at depth 4.

For each pair, all component records retain nonempty title, author, venue, and year fields as non-constrained distractor assertions. Each gold edge is encoded as a unique exact key value shared by its endpoints. The runner adds one controlled single-valued predicate: unequal values give a positive and equal values give its paired negative. Across 136 instances, CWI returns all 68 positive witnesses, rejects all 68 negatives, and every witness is rechecked against source assertions. Components range from 2 to 37 records; contexts from 12 to 224 triples; natural distractors from 6 to 113; and positive witnesses from 6 to 15 triples. The labels validate the

| Triples | Entries | CWI $q=1$ | Fresh $q=1$ | CWI $q=100$ | Fresh $q=100$ |
|---------|---------|-----------|-------------|-------------|---------------|
| 1,689   | 1,386   | 0.69      | 1.07        | 1.19        | 111.26        |
| 16,809  | 13,806  | 9.77      | 20.02       | 10.39       | 1,732.23      |
| 56,009  | 46,006  | 39.79     | 75.19       | 40.31       | 7,505.44      |

Table 3: Median total milliseconds. “Entries” counts structural index items plus materialized conflict pairs; it is not a heap-byte estimate.

| $n$ | Triples | Pairs  | Ingest ms, median [IQR]    |
|-----|---------|--------|----------------------------|
| 8   | 24      | 28     | 0.095 [0.059, 0.107]       |
| 16  | 48      | 120    | 0.336 [0.289, 0.555]       |
| 32  | 96      | 496    | 0.505 [0.382, 0.587]       |
| 64  | 192     | 2,016  | 3.014 [2.751, 3.370]       |
| 128 | 384     | 8,128  | 19.266 [18.584, 20.006]    |
| 256 | 768     | 32,640 | 253.058 [245.788, 264.722] |

Table 4: Median [IQR] over 11 runs after one warm-up when all records share one key and have distinct constrained values.

retrieval mechanism on natural topology, not the prevalence of semantic conflicts.

### Maintenance Workload Matrix

For each of 300, 3,000, and 10,000 persons, the generator and schema parser run before timing. One unrecorded warm-up precedes 11 measured repetitions. In a repetition, CWI ingests the in-memory snapshot once and answers the first  $q \in \{1, 10, 100\}$  deterministic conflict queries. The stateless arm receives the same in-memory triples but reconstructs its triple index, exact key closure, and conflict cells for every query. Thus neither arm receives a file-I/O advantage.

Every query has exactly one expected conflict and no run is missing or spurious. Full per-run timings and IQRs are shipped. The retained-item ratio is 0.820–0.821 per source triple on this generator; the fresh arm retains no derived state between queries.

### Large-Class Stress Matrix

This adversarial leg measures the disclosed worst regime rather than store-size scaling: the maintained pair set is exactly  $n(n-1)/2$ . Timings are descriptive and machine-dependent; pair and triple counts are deterministic.

### Artifact Contract

The anonymous code/data archive contains:

- TypeScript source and focused tests for the maintained index, exact comparators, heuristic and exact joint-budget queries, adaptive and on-demand witnesses, workload comparison, retrieval policies, instance scoring, and scale ledger;
- all eight synthetic fixture families and their instance manifests;
- frozen paper-facing outputs for exact, maintained, on-demand, sparse, dense, curve, joint-budget, semi-natural, workload, class-stress, and scale analyses;

- the pinned Python environment and runner for the dense baseline; and
- the external topology and semi-natural scripts with frozen reports.

The archive deliberately omits repository history, internal planning documents, hosted-reader runs excluded from the paper, service credentials, caches, installed dependencies, compiled output, and submission-production files. Dense re-execution requires obtaining the cited third-party encoder weights; frozen dense outputs are included so the reported table remains auditable without that download. Timing will vary by machine and should be regenerated only as a systems check, not expected to match byte for byte. For double-blind review, the project-owned RDF namespace is replaced consistently in source, fixtures, and frozen textual outputs by a reserved `example.org` namespace. This name-only substitution changes no graph structure, label, measurement, or scientific value.

### **Additional Limitations**

Independent shortest-witness selection need not minimize the joint union; exact selection resolves this only when exponential path enumeration and branch-and-bound finish below their caps. Materialized unequal-value pairs can be quadratic within a large equivalence class; the dedicated stress leg exposes that regime. The semi-natural conflict predicate is injected, and unequal natural publication-year strings remain unlabeled because versions, linkage errors, or metadata errors could explain them. The workload uses a single process and an insertion-batch/read-many schedule, not concurrent updates, deletion, or heap-residency measurement.